



Day 1: The Foundations of Data & AI (NumPy & Pandas)

Series: Zero to AI/ML Engineer 30-Day Roadmap

Target Goal: Master data structures and manipulation—the literal fuel for all Machine Learning algorithms.



1. Why Are We Starting Here?

Before you can build complex Neural Networks or Train Large Language Models, you must feed them data. Real-world data is dirty, unorganized, and massive.

- **NumPy** handles the heavy mathematical multi-dimensional arrays (vectors and matrices).
- **Pandas** wraps around NumPy to give you "DataFrames" (structured tables, just like

Excel sheets, but powered by Python code).



2. NumPy: Multi-Dimensional Arrays

NumPy (Numerical Python) is optimized for performance because its arrays are stored in contiguous memory blocks, making computation vastly faster than standard Python lists.

Core Concepts to Learn Today:

1. **Array Creation:** Converting a Python list into a high-performance vector.
2. **Shape & Reshaping:** Understanding the dimension of your data (e.g., changing a 1\times100 vector into a 10\times10 matrix).
3. **Vectorization:** Performing operations on an entire array instantly without using slow for loops.

Key Syntax Cheat Sheet:

```
import numpy as np
```

1. Create a 1D Array (Vector)

```
vector = np.array([1, 2, 3, 4, 5])
```

2. Create a 2D Array (Matrix - standard format for ML features)

```
matrix = np.array([[1, 2, 3], [4, 5, 6]])
```

3. Generate placeholder arrays (Crucial for weights initialization in Deep Learning)

```
zeros_array = np.zeros((3, 3)) # 3x3 matrix of 0s
```

```
random_array = np.random.rand(2, 2) # 2x2  
matrix of random numbers between 0 and 1
```

4. Matrix Math (Fast Execution)

```
squared_vector = vector ** 2 # Instantly squares  
every element: [1, 4, 9, 16, 25]
```



3. Pandas: Data Manipulation & Wrangling

Pandas takes your NumPy arrays and converts them into structured tabular data where columns have names and rows have indices. This is where you will spend 60% of your time as a Data Scientist.

Core Concepts to Learn Today:

1. **DataFrames:** Understanding rows, columns, and indexes.
2. **Data Loading:** Pulling data into Python from external files (CSV, Excel, JSON).
3. **Filtering & Inspection:** Viewing parts of your data and filtering rows based on logic conditions.

Key Syntax Cheat Sheet:

```
import pandas as pd
```

1. Create a DataFrame from scratch

```
data = {  
'Student_ID': [101, 102, 103],  
'ML_Score': [95, 88, 92],  
'Completed_Projects': [3, 1, 4]
```

```
}
```

```
df = pd.DataFrame(data)
```

2. Loading real-world data (How you will actually do it)

```
df =
```

```
pd.read_csv("your_dataset.csv")
```

3. Inspecting the Data

```
print(df.head(2)) # View the first 2 rows
```

```
print(df.info()) # Check for data types and  
missing (null) values
```

```
print(df.describe()) # Get statistical summary  
(mean, median, standard deviation)
```

4. Filtering Data (Conditional Selection)

Find all students who scored above 90 in ML

```
top_students = df[df['ML_Score'] > 90]  
print(top_students)
```

Day 1 Action Items (Your Homework)

To actually learn this, stop watching tutorials and write the code yourself:

1. Set up a fresh file in VS Code named `day1_practice.py`.
2. Generate a 1D NumPy array containing numbers from 1 to 20, and reshape it into a

4x5 matrix.

3. Find a free, public dataset on **Kaggle** (like the *Titanic Dataset* or *Iris Dataset*), download the CSV file, load it using Pandas, and print out its statistical summary (.describe()).